

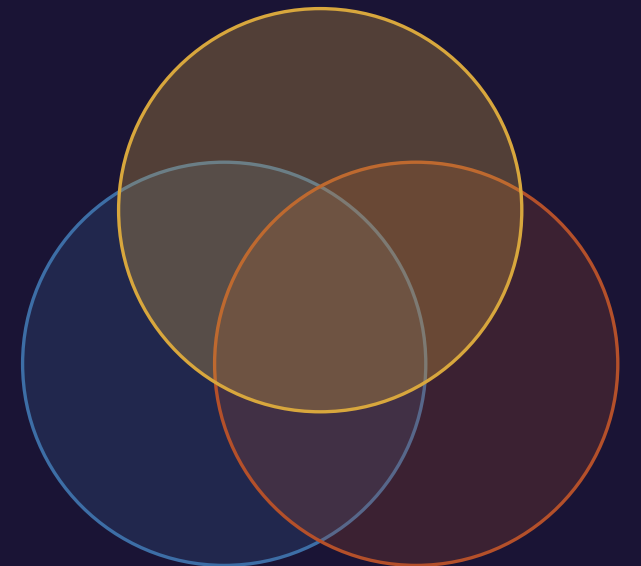
A LESSON IN RESEARCH METHODOLOGY

The Dialectic of Research in Computer Science

Models of Argument: How Claims Are Made, Contested, and Defended

From Socratic dialectic and philosophy of science to the Toulmin model —
and how both structure the anatomy of a CS research paper.

Research Methodology in CS



A Paper Is an Argument, Not Just an Artifact

- 1 A paper doesn't just report what you built — it argues that a claim about it is true and matters.
- 2 “Results” sections are evidence. “Related work” pre-empts rebuttal. “Limitations” is a qualifier.
- 3 Reviewers rarely evaluate your artifact directly — they evaluate the argument you built around it.
- 4 Two teams can build the same system and be judged very differently on argument quality alone.

THE CRITICAL-RATIONALIST FRAMING

Scientific knowledge does not advance by accumulating proofs. It advances because claims are put forward boldly — and then survive serious attempts to refute them.

— a paraphrase of Popper's critical rationalism

If this is true of a field, it is true of a single paper: your claim only earns belief once it has faced — and survived — its own counter-arguments.

What Is a Dialectic?

Dialectic (n.): a method of reasoning that proceeds by exchanging opposing arguments, in the belief that truth emerges from the *collision* of claim and counter-claim — not from one voice alone.

Socrates used it as a method of dialogue — cross-examining a claim until its hidden assumptions broke down. Hegel later formalized it as a three-step engine of intellectual progress:



This isn't a museum piece — it's the skeleton of how a research field, and a single research paper, actually argues its way forward.

How Does Scientific Knowledge Actually Move Forward?

Popper

FALSIFICATIONISM

Bold conjectures are put forward, then hunted for decisive counter-examples. A single clean refutation outweighs any number of confirmations.

CS example: a benchmark that refutes “algorithm X always beats Y” kills the universal claim outright.

Kuhn

PARADIGM SHIFTS

“Normal science” solves puzzles inside a shared paradigm until anomalies pile up and trigger a revolutionary, hard-to-compare shift.

CS example: the shift from hand-built symbolic AI to statistical / connectionist learning as a paradigm change.

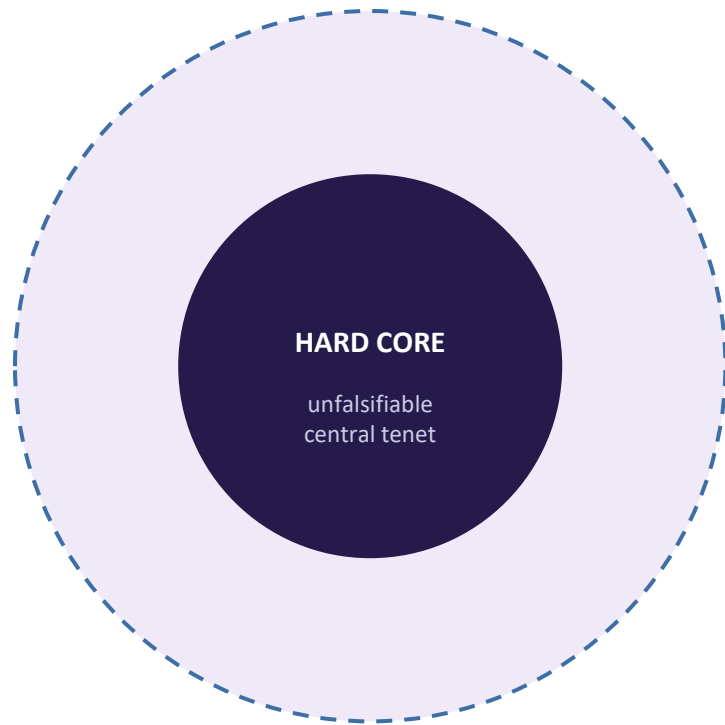
Lakatos

RESEARCH PROGRAMMES

A protected “hard core” is shielded by a flexible “protective belt” of auxiliary hypotheses that absorb anomalies.

CS example: deep learning as a programme — architectures and tricks change; the core commitment doesn't.

Lakatos in Practice: Hard Core & Protective Belt



PROTECTIVE BELT — auxiliary hypotheses, absorbing anomalies

Negative heuristic: protect the hard core — redirect anomalies into the belt instead.
Positive heuristic: a roadmap for extending the programme and anticipating future problems.

PROGRESSIVE

New auxiliary hypotheses predict genuinely novel facts — later confirmed.
Example: deep learning → scaling laws, predicted and then observed.

DEGENERATING

Changes are only ad hoc patches with no new predictive power. The programme survives by explaining away failures after the fact.

PART 1 → PART 2

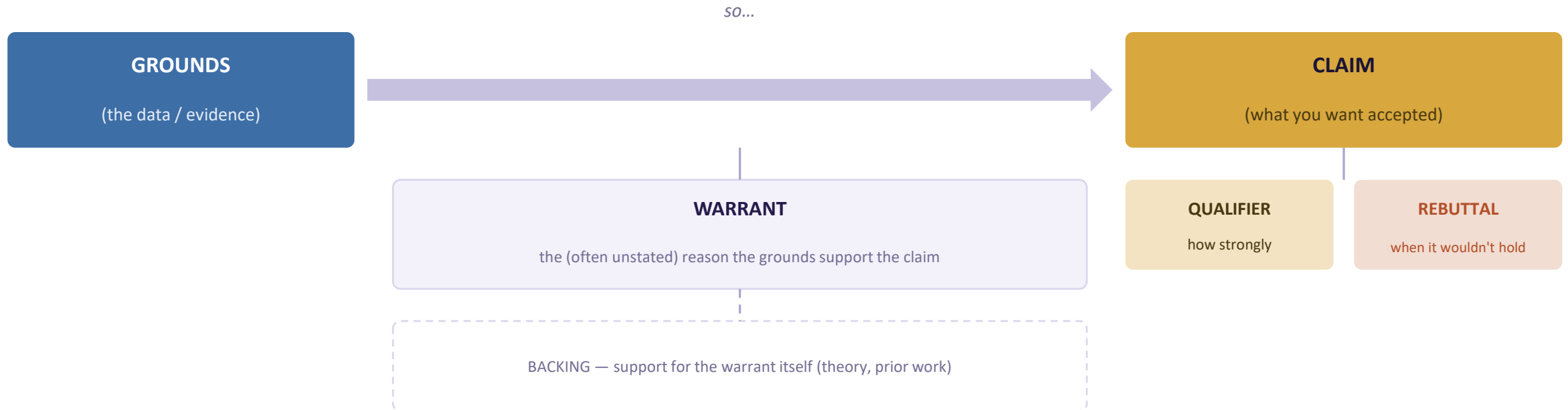
From the philosophy of a field to the argument on the page

If knowledge in a field advances dialectically — claim meeting counter-claim — how is a single argument, for a single claim, actually built and defended on paper?

Next: a working model for the anatomy of one argument — Toulmin's model.

The Toulmin Model of Argument

Stephen Toulmin (*The Uses of Argument*, 1958) proposed a practical, layout-based model of argument as an alternative to purely formal logic — now standard in composition and STEM writing pedagogy.



The Six Components, Defined

1

Claim

The central assertion you want your reader to accept — a belief or a call to action.

2

Grounds (Data)

The facts, measurements, or evidence offered as direct support for the claim.

3

Warrant

The logical bridge explaining why the grounds actually support the claim — often implicit.

4

Backing

Further support for the warrant itself, when the bridge needs its own justification.

5

Qualifier

Words that temper the claim's strength: “most,” “typically,” “under condition X.”

6

Rebuttal

The conditions under which the claim would not hold — named in advance.

Mapping Toulmin onto a CS Paper

TOULMIN COMPONENT	WHERE IT LIVES IN A CS PAPER
Claim	Contribution statement — abstract & introduction
Grounds	Experimental results, benchmarks, proofs, measurements
Warrant	The methodological / statistical reasoning linking results to claim — often left implicit
Backing	Related theory, validated prior techniques, citations
Qualifier	Scope statements: “for graphs of bounded degree...,” “on workloads similar to...”
Rebuttal	Threats-to-validity / limitations section

Worked Example: A Systems Paper

Claim under test: “Our caching policy X reduces P99 tail latency by 30% over LRU.”

Grounds	Benchmark results across 3 production traces, measured against an LRU baseline.
Warrant	These traces are representative of real workloads, so the latency reduction generalizes.
Backing	The traces are established public benchmarks used by prior published caching work.
Qualifier	Holds for read-heavy workloads with skewed key-access distributions.
Rebuttal	May not hold under write-heavy or adversarial access patterns — addressed in §Limitations.

What Counts as a Result?

Mary Shaw's taxonomy of research contributions in software engineering / CS (ICSE 2003; revisited 2017)

1

Procedure / Technique

A new or better way to do a task — design, implementation, measurement — operational and reproducible.

2

Qualitative / Descriptive Model

Taxonomies, architectural styles, design patterns — structural frameworks for organized observation.

3

Empirical Model

A predictive model grounded in data observed from real usage or controlled studies.

4

Analytic Model

A formal, mathematical structure enabling rigorous analysis or automated reasoning.

5

Tool / Notation

An implemented tool or formal language with computational foundations.

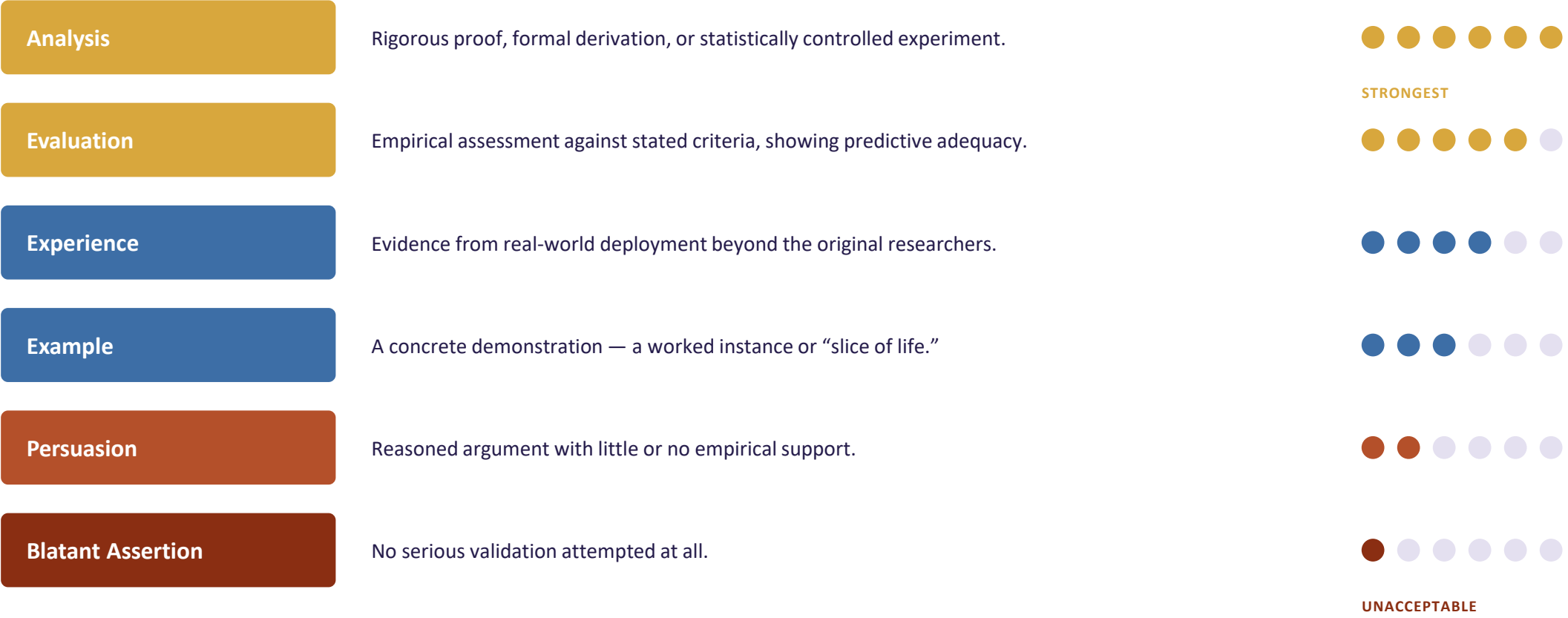
6

Specific Solution / Prototype

Direct application of engineering principles to one particular problem.

What Counts as Evidence?

Shaw's validation types, ordered by evidentiary strength — this is what “Grounds” must actually be made of.



Beyond Toulmin: Argumentation Schemes

Toulmin gives the skeleton of one argument. Walton, Reed & Macagno catalog recurring, defeasible patterns of everyday reasoning — each paired with critical questions that can defeat it.

1

Argument from Expert Opinion

“Cited authority E asserts P, so P is plausible.”

CRITICAL QUESTIONS

Is E actually an expert in this sub-domain? Did the cited result measure the same thing you're claiming?

2

Argument from Example

“The system worked on these instances, therefore it generalizes.”

CRITICAL QUESTIONS

Are the examples representative? Is there a plausible counter-example the paper didn't test?

3

Practical Reasoning

“We should adopt X because it achieves goal G.”

CRITICAL QUESTIONS

Are there other ways to achieve G? What side effects does adopting X bring?

Every Contribution Is a Small Dialectic



WORKED EXAMPLE

THESIS

Symbolic planners guarantee correctness but don't scale.

ANTITHESIS

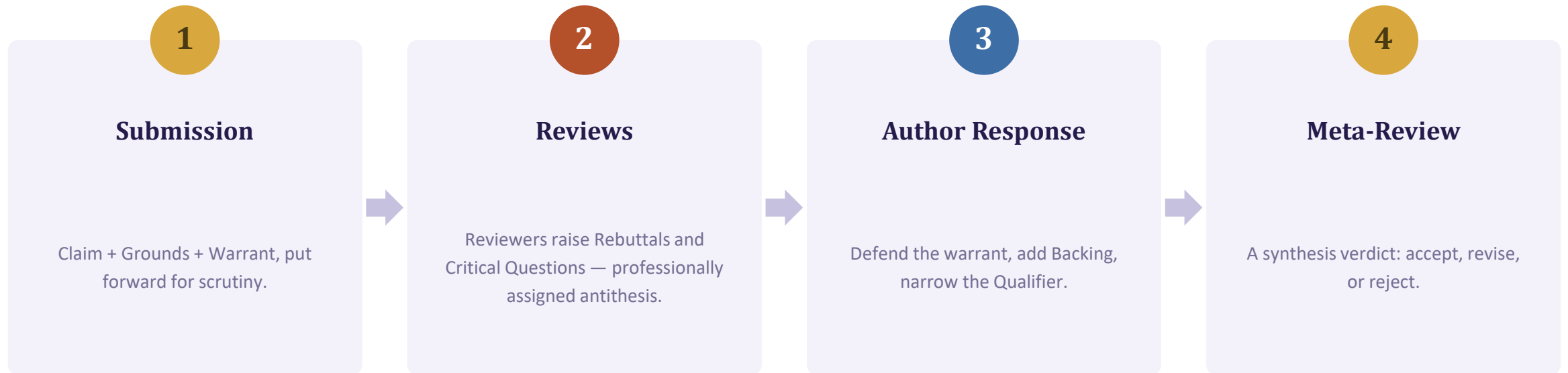
Learned policies scale but offer no guarantees.

SYNTHESIS

Neuro-symbolic methods pair learned heuristics with symbolic verification.

This is why “Related Work” reads best as antithesis-building, not a summary of prior papers.

Peer Review: The Dialectic, Made Procedural



Rebuttal periods exist precisely because the model predicts they must: a claim that has not yet met its antithesis has not yet been tested.

Why Every Paper Needs a Limitations Section

Omitting Toulmin's rebuttal component doesn't remove the counter-arguments — it just leaves them unaddressed for a reviewer to raise instead, on their terms.

WEAK

“This work has some limitations and future work should address them.”

Generic hedging. Names no specific condition, so it pre-empts nothing — a reviewer still has to do the work of finding the real weak point.

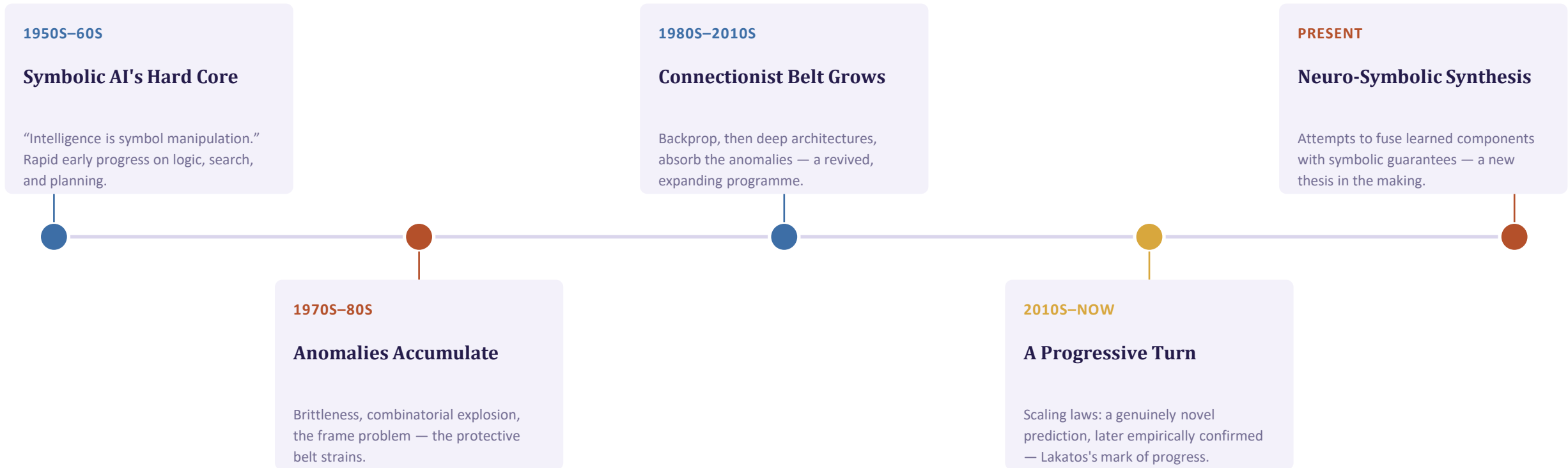
STRONG

“Results hold for skewed, read-heavy access patterns; under write-heavy or adversarial patterns, we expect the gain to shrink (§6.3).”

Specific scope + named condition + pointer to evidence. This is the Rebuttal component actually doing its job.

Case Study: A Field-Level Lakatosian Contest

Symbolic AI vs. connectionism, read as competing research programmes



Argumentative Fallacies to Watch For

Cherry-Picked Benchmarks

Selecting only the datasets or baselines that favor your method.

Moving the Goalposts

Redefining the success metric after seeing which results looked good.

Appeal to Novelty

“No one has tried this before” offered as the sole justification.

Hasty Generalization

One dataset or domain, generalized into a universal claim.

Unfalsifiable Framing

Stating the claim so no possible result could count against it.

Post-Hoc Warrant

Building the “why it works” story only after seeing favorable results.

A Checklist for Strengthening Your Argument

-  State your claim precisely, with an explicit qualifier that bounds its scope.
-  Make your warrant explicit — don't leave the “why does this evidence support this” step for the reader to supply.
-  Back your warrant: cite the theory or validated method that licenses your inferential leap.
-  Actively hunt for rebuttals — what result would make this claim false, and did you go look for it?
-  Match validation type to claim type (Shaw) — a qualitative model needs evaluation or experience, not just an example.
-  Write “related work” as antithesis-building, not a summary — name the tension your work resolves.

Discussion

- 1 Pick a paper you've read recently. Reconstruct its Toulmin structure — what's the warrant, and is it explicit or hidden?
- 2 By Lakatos's criteria, is your own thesis or project's research programme progressive or degenerating right now? What would a novel, confirmed prediction look like for it?
- 3 Find a claim in a CS paper with no rebuttal addressed. What's the strongest antithesis a reviewer could raise against it?
- 4 Where does rigorous “empirical” work end and Shaw's weak sense of “persuasion” begin, in your own subfield?

Key Takeaways

- 1 Research knowledge advances dialectically — claims meeting counter-claims (Popper, Kuhn, Lakatos) — not by accumulation alone.
- 2 A CS paper is itself a Toulmin argument: claim, grounds, warrant, backing, qualifier, and rebuttal.
- 3 The weakest link is usually the warrant — the unstated reasoning connecting your evidence to your claim.
- 4 Different claim types demand different validation (Shaw) — match your method to the type of result you're making.
- 5 Treat related work as antithesis and limitations as institutionalized rebuttal: that's what turns a report into an argument.

Further Reading

- **Toulmin, S. E.** *The Uses of Argument*. Cambridge University Press, 1958.
- **Popper, K.** *The Logic of Scientific Discovery*. Hutchinson, 1959 (Ger. orig. 1934).
- **Kuhn, T. S.** *The Structure of Scientific Revolutions*. University of Chicago Press, 1962.
- **Lakatos, I.** *The Methodology of Scientific Research Programmes*. Cambridge University Press, 1978.
- **Walton, D., Reed, C., & Macagno, F.** *Argumentation Schemes*. Cambridge University Press, 2008.
- **Shaw, M.** *Writing Good Software Engineering Research Papers*. Proc. ICSE, 2003.
- **Theisen, C., et al.** *Writing Good Software Engineering Research Papers: Revisited*. Proc. ICSE-C, 2017.